

A Lightweight Selective-Routing Study for CPU-Only Inference

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Transparency and Provenance Note

This document is an early research draft built around a simple personal idea: a lightweight inference system may benefit from keeping a small always-on guard model active for routine inputs while selectively activating stronger experts only for difficult cases. The author does not claim prior expertise in multi-agent inference or conditional-computation theory and presents this work as an exploratory experiment rather than a settled scientific result.

Parts of the brainstorming, code scaffolding, analysis, and manuscript drafting were produced with the assistance of large language models, including ChatGPT, Gemini, and Grok, and were then selected, checked, and edited by the author. The main purpose of including this note is transparency: the underlying idea came from the author, while LLMs were used as exploratory and writing tools to help turn that idea into runnable experiments and a readable draft.

For additional transparency, the author provides the originating LLM conversations and the experiment notebook used for the experiments.

Summarized ChatGPT conversation

<https://chatgpt.com/share/69e3d699-ff28-8320-8e8a-1a427d369309>

Originating Gemini conversation

<https://gemini.google.com/share/2084e6d80e64>

Originating Grok conversation

https://grok.com/share/bGVnYWN5LWNvcHk_47c2231a-c875-4cad-8deb-e06b94c4763b

Experiment notebook / Colab

https://colab.research.google.com/drive/1_4zEYPd5MedGnjoAcRdDK-0yDEUHsSYN?usp=sharing

This manuscript should not be interpreted as a peer-reviewed claim of novelty, priority, or superior performance. Readers should independently verify experiments, compare against prior work, and treat the present results as an initial proof-of-concept study.

Abstract

This paper studies a lightweight selective-routing framework for CPU-only inference on resource-constrained devices. The system keeps a small always-on guard active for routine inputs and selectively activates stronger experts only when a routing signal indicates difficulty. Rather than introducing a new model family, the work studies a model-agnostic routing policy built around a guard, an expert stage, and a cost-aware trigger.

The main empirical finding is regime-dependent. In balanced settings such as digits, margin is the strongest default routing signal and combined routing can slightly improve the accuracy–cost frontier by adding a secondary notion of fragility. In the final real credit-card-fraud benchmark, however, the best deployable policy is extremely conservative: under a 0.1% trigger-budget cap, wobble preserves guard-level performance with mean balanced accuracy 0.9288, mean positive recall 0.8606, and mean trigger rate 0.0006.

Overall, the paper characterizes when selective routing helps and when it should stay almost silent. The results suggest that uncertainty signals dominate in balanced regimes, while stability cues become most valuable when escalation must remain extremely rare.

1 Introduction

This paper explores a simple selective-inference idea: instead of relying on a single model for every input, the system keeps a lightweight always-on controller and activates additional experts only when the task appears harder. In this view, easy inputs are handled cheaply by the guard model, while harder inputs trigger a short-lived burst of stronger computation. After the burst, the system returns to a low-cost mode, similar to a temporary burst of extra computation followed by a return to the low-cost path.

The central intuition is that not every example deserves the same amount of computation. A guard can monitor confidence, trigger extra experts when uncertainty rises, aggregate their predictions, and then deactivate them once the difficult region has passed. This creates a dynamic pipeline with four ingredients: (i) an always-on guard, (ii) an expert stage that is invoked only when needed, (iii) a routing signal that estimates hardness, and (iv) an explicit cost trade-off that prevents constant escalation. Rather than redesigning models themselves, the paper studies a lightweight orchestration layer that can wrap existing small classical models.

The main question is therefore not whether stronger experts can improve accuracy in principle, but under what conditions a tiny routing policy can improve the accuracy–efficiency trade-off relative to strong always-on baselines.

Contributions. This work makes three concrete contributions:

- It develops a lightweight, CPU-only selective routing framework that dynamically switches between a guard model and stronger experts on a per-sample basis.
- It evaluates margin, wobble, combined, confidence, random, always-expert, and oracle baselines under repeated-seed and budget-constrained protocols.
- It shows that routing is regime-dependent: margin is the strongest default signal in balanced settings, while wobble is the safest ultra-low-budget trigger under extreme class imbalance.

2 Related Work

This work sits between conditional computation, selective prediction, and resource-aware inference. Related ideas include early-exit networks, mixture-of-experts routing, abstention and deferral systems, and adaptive computation methods. The present paper differs in emphasis: it keeps the

models simple, studies CPU-only classical learners rather than large neural networks, and treats the routing controller itself as the main object of interest.

A second connection is to uncertainty estimation and calibration. Confidence, entropy, and margin are standard tools for deciding whether a model should act or defer. Here, that idea is extended with a stability-oriented signal, wobble, which measures whether the model’s prediction changes under small perturbations. The main empirical lesson is that wobble is usually not a replacement for margin, but it can be a useful modifier when escalation must remain very rare.

3 Method Overview

The proposed system uses a small guard model to score every input first. If the routing score stays below a threshold, the guard prediction is returned directly. If the score exceeds that threshold, the system activates a stronger expert stage and then falls back to low-cost inference on the next sample. In this sense, the method is best understood as a tiny control policy for selective expert activation rather than as a new predictive architecture.

Let $g(x)$ denote the guard prediction for input x , let $e(x)$ denote the prediction returned by the stronger expert stage, and let $s(x)$ denote a hardness score. The routing policy is

$$\hat{y}(x) = \begin{cases} e(x), & s(x) > \tau, \\ g(x), & \text{otherwise,} \end{cases}$$

where τ is selected on validation data. The experiments study several forms of $s(x)$, including raw margin, wobble-based instability, and a combined score that fuses uncertainty and fragility.

3.1 Cost-Aware Objective

The system is trained and selected with a cost-sensitive validation objective. If $T(x) \in \{0, 1\}$ is the trigger variable and $P(T(x) = 1)$ is the observed trigger rate, then threshold selection maximizes

$$\text{score} = \text{accuracy} - \lambda \cdot P(T(x) = 1),$$

where $\lambda > 0$ penalizes expensive escalation. This prevents the degenerate solution in which the router simply calls the expert on almost every sample.

3.2 Routing Signals

We evaluate three routing signals that estimate whether a sample should be escalated to the expert stage.

Margin. Margin is defined as the difference between the guard’s top two predicted class probabilities. Low margin indicates ambiguity and is treated as a signal of difficulty.

Wobble. Wobble measures prediction instability under small input perturbations. For each sample, repeated noisy perturbations are applied and the fraction of prediction flips is recorded. Higher wobble indicates greater fragility.

Combined Signal. To capture both ambiguity and fragility, the combined score fuses normalized margin and wobble into a single hardness signal:

$$S_{\text{combined}}(x) = \alpha \cdot (1 - \text{margin}_{\text{norm}}(x)) + (1 - \alpha) \cdot \text{wobble}_{\text{norm}}(x),$$

with $\alpha = 0.65$ in the reported experiments.

4 Experimental Progression

This section summarizes the experimental path that led from the first prototype to the final routing benchmark. The full tables, per-setting plots, and supporting diagnostics are collected in the appendix so that the main paper stays focused on the core routing lesson: the value of selectivity depends on the regime.

The appendix contains the full tables, plots, and seed-level diagnostics; the main text reports only the central takeaway from each stage.

4.1 Prototype Stage: Heart Disease

The first prototype on the Cleveland heart disease dataset established the basic guard–expert loop, but it also showed the first failure mode: weak helpers can reduce latency without preserving predictive quality. These results are kept mainly as a diagnostic starting point; the full numbers are moved to Appendix A.

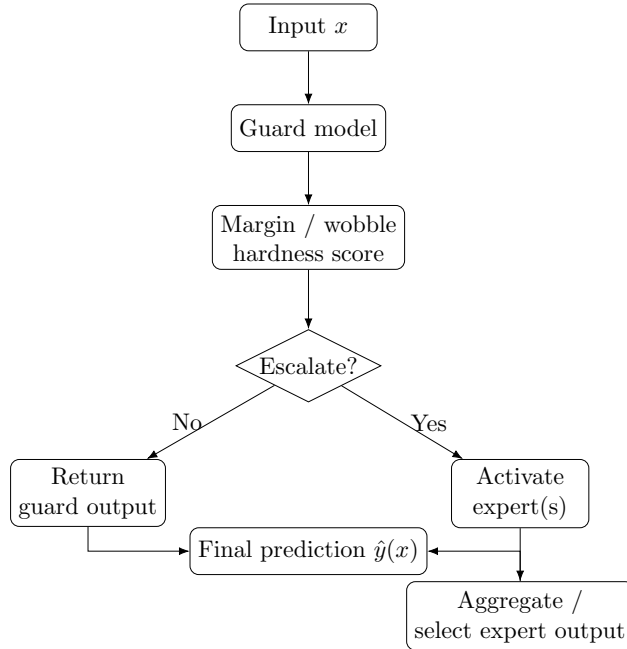


Figure 1: Hybrid burst orchestration pipeline. A lightweight guard processes every input and triggers stronger experts only for cases that appear difficult under the routing signal.

4.2 Digits Stage: When Routing Becomes Competitive

The digits dataset provided the first setting where routing became genuinely competitive. After strengthening the burst stage, the expert pathway became competent, disagreement became rare, and the guard stress test showed that the guard was the main bottleneck. This made digits the cleanest place to study routing selectivity. The detailed diagnostics are in Appendix A.4.

4.3 Signal Study: Margin, Wobble, and Combined Routing

The signal-ablation study isolated the routing rule itself. Margin remained the strongest general-purpose trigger, wobble captured perturbation fragility, and the combined score slightly reshaped the frontier in the balanced operating region. The full Pareto tables are in Appendix A.5.

4.4 Cross-Dataset Check

A simple generalization test on digits, wine, and breast cancer showed that routing behavior depends on guard headroom. Digits offered the clearest benefit, while wine and breast cancer were already close to solved by the guard. The detailed budget curve is in Appendix A.6.

4.5 Final Rare-Event Benchmark

The final benchmark used the real credit card fraud dataset to test the controller under extreme class imbalance. The important result was not a dramatic gain over the guard, but that wobble was the safest ultra-low-budget trigger: when escalation had to remain rare, wobble preserved guard-level performance while avoiding unnecessary expert calls. Full results are in Appendix A.7.

5 Discussion and Lessons Learned

The main conceptual lesson is simple: the architecture works as a division of labor. A small guard handles routine cases, and stronger experts are used only when the routing signal suggests that an input is genuinely difficult. That makes the system a routing layer rather than a new predictive model, which is the right way to interpret the contribution.

The second lesson is that different signals matter in different regimes. In balanced tasks, margin remains the strongest general-purpose trigger because it captures uncertainty near the decision boundary. Wobble is weaker as a standalone signal, but it captures a different property — fragility under perturbation — and therefore becomes useful as a modifier that can improve the balanced part of the frontier. This is why the combined signal is interesting: it does not replace margin, but it can reshape the trade-off curve.

The third lesson is that routing is regime-dependent. When the guard already leaves substantial headroom, selective escalation can recover most of the expert benefit at lower cost. When the guard is already strong and the budget is extremely tight, as in the fraud benchmark, the best policy is often to stay almost silent. That boundary case is valuable because it shows the method is not universally beneficial; it is beneficial only when the guard, expert, and budget regime are well matched.

These results suggest a clear next step: instead of adding more signals for their own sake, future work should focus on improving selectivity, calibration, and budget control. The appendix contains the full tables and plots, while the main text keeps only the core takeaways.

6 Limitations and Future Work

This work has several limitations. First, the experiments are conducted on relatively small datasets and classical machine-learning models rather than on real embedded deployments with direct power measurements. Second, routing gains depend strongly on the gap between guard and expert performance; when experts are only marginally better, selective escalation offers limited benefit. Third, even the best routing signals can still over-trigger under some settings, indicating that calibration and explicit trigger-budget constraints remain open problems.

Future work should therefore focus on stronger hardness predictors, stricter cost constraints, better guard calibration, and evaluation on genuine low-resource hardware.

7 Conclusion

The experiments in this paper support a simple but useful conclusion: a tiny CPU-only routing controller can improve the accuracy–efficiency trade-off, but only when the guard, expert, and trigger are well matched to the task. Raw margin is the best default signal, wobble is a useful stability cue, and the combined signal is the most interesting frontier-improving variant. The architecture is therefore viable as a lightweight orchestration layer, not as a new model family. Its real value is in showing when to spend computation, when to save it, and how to make that choice in a way that is interpretable, repeatable, and cheap enough to matter.

A Appendix: Full Experimental Details

A.1 Heart Disease MVP Results

The first prototype compared FP32, quantized-style, pure burst, and hybrid inference on the Cleveland heart disease dataset. It was useful as a diagnostic stage because it showed that weak helpers can hurt predictive quality even when they reduce computation.

Table 1: Heart disease prototype summary.

Method	Accuracy	Macro F1	Wall ms/sample	Trigger rate
FP32 baseline	0.877	0.8751	18.4064	–
Quantized-style wrapper	0.877	0.8751	13.4444	–
Pure Burst	0.833	0.8282	167.7704	0.487
Hybrid (quantized guard + burst)	0.821	0.8116	131.5921	0.387

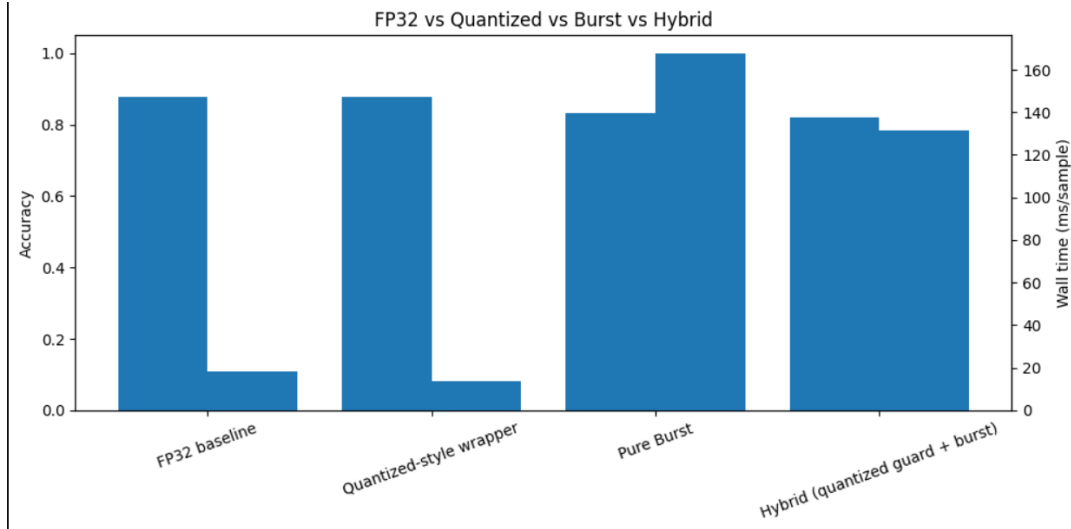


Figure 2: Heart-disease prototype comparison used during the earliest development stage.

A.2 Light v2 Follow-Up

The lighter follow-up confirmed the same overall lesson: the control logic worked, but the helper stage remained too weak to outperform the baseline in an accuracy-first setting.

Table 2: Light v2 follow-up summary.

Method	Accuracy	Runtime (s)	Peak mem. (KB)	Notes
Baseline	0.843	13.82	351.2	Reference model
Quantized	0.843	9.45	348.2	Same accuracy, lower time
Pure Burst	0.777	15.06	358.4	Disagreement = 0.229
Hybrid	0.744	6.03	360.2	Disagreement = 0.252

A.3 Light v3 Digits Results

The stronger digits setup made the routing story much clearer. The burst pathway became competent, disagreement became rare, and the routing frontier became meaningful.

Table 3: Light v3 digits summary.

Method	Accuracy	Macro F1	Wall ms/sample	Trigger rate	Burst acc.
FP32 full	0.9611	0.9607	80.6982	—	—
Quantized-style full	0.9611	0.9607	71.5888	—	—
Pure Burst	0.9111	0.9090	42.0769	0.3000	0.9352
Hybrid	0.9083	0.9064	42.0267	0.3000	0.9352

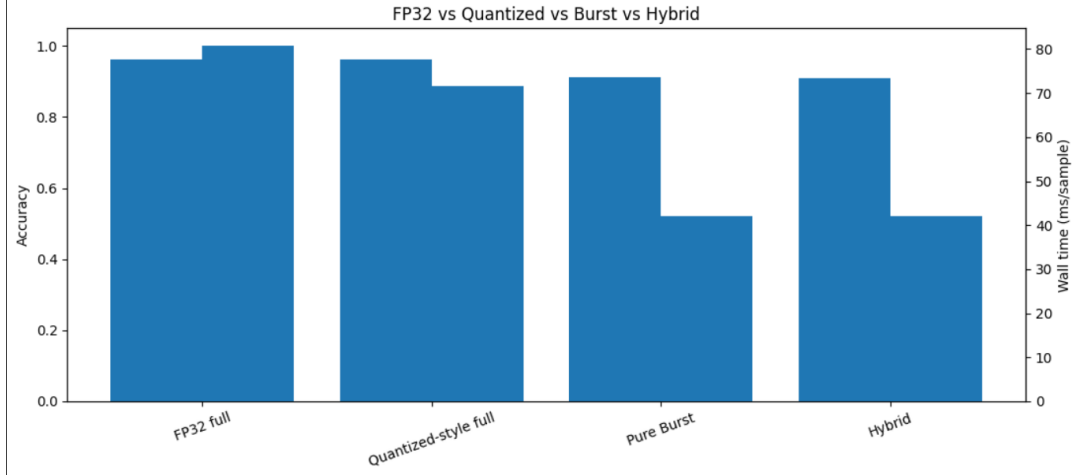


Figure 3: Light v3 digits comparison showing the stronger expert pathway and the resulting routing trade-off.

A.4 Guard Stress Test Diagnostics

The guard stress test showed that the guard was underconfident rather than overconfident, and that the high-recall operating point came at the cost of aggressive escalation.

Table 4: Digits guard stress test summary.

Setting	Accuracy	Confidence / ECE	Trigger rate	Cost
Guard only	0.8639	0.4681 / 0.3958	—	1.00
Expert only	0.9611	—	1.0000	9.00
Guard + expert at threshold 0.50	0.9611	—	0.6083	5.87

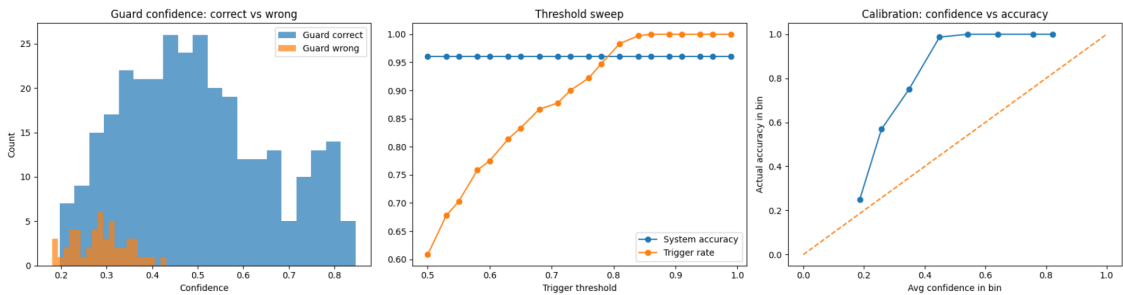


Figure 4: Guard stress-test diagnostics referenced in the main text.

A.5 Signal Ablation and Pareto Frontier

Margin remained the strongest default trigger, wobble remained the weakest standalone signal, and the combined score was mainly useful as a frontier modifier.

Table 5: Digits Pareto frontier summary.

Routing Signal	λ Penalty	Mean Acc	Trigger %	Hard Recall	Cost (Units)
Wobble (Low Noise)	0.001	0.9078	3.06%	0.1240	1.24
Wobble (High Noise)	0.001	0.9378	13.44%	0.5024	2.07
Margin (Standard)	0.080	0.9706	33.72%	0.9411	3.69
Combined (Balanced)	0.050	0.9717	38.39%	0.9526	4.07
Combined (Max Acc)	0.001	0.9756	49.17%	0.9952	4.93
Always-On Expert	—	0.9711	100%	—	8.00

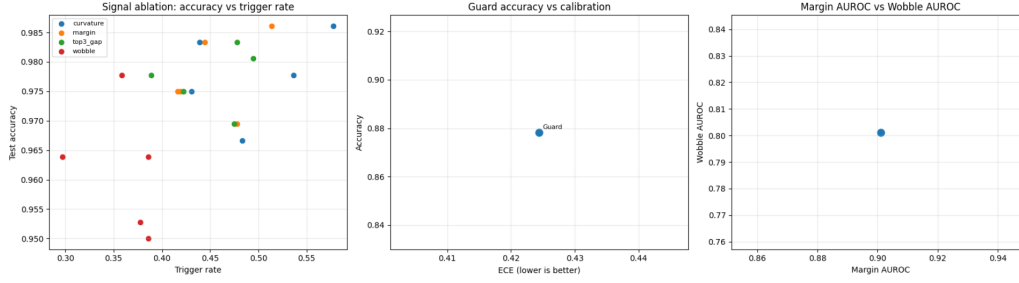


Figure 5: Signal-ablation diagnostics and trade-off visualization for the digits routing study.

A.6 Cross-Dataset Generalization

Table 6: Cross-dataset routing summary at roughly 40% expert usage.

Dataset	Guard acc.	Expert acc.	Routed acc.	Trigger rate
Breast cancer	0.9518	0.9591	0.9591	0.3830
Digits	0.8266	0.9745	0.9550	0.3996
Wine	0.9769	0.9815	0.9815	0.3981

Table 7: Budget curve across datasets.

Dataset	Budget	Accuracy	Trigger
Cancer	0.1	0.9605	0.1009
Cancer	0.2	0.9591	0.1988
Cancer	0.3	0.9591	0.2997
Cancer	0.4	0.9591	0.3830
Cancer	0.5	0.9591	0.4371
Digits	0.1	0.8744	0.1001
Digits	0.2	0.9124	0.1980
Digits	0.3	0.9314	0.3004
Digits	0.4	0.9550	0.3996
Digits	0.5	0.9634	0.4979
Wine	0.1	0.9861	0.1111
Wine	0.2	0.9815	0.2083
Wine	0.3	0.9815	0.3056
Wine	0.4	0.9815	0.3981
Wine	0.5	0.9815	0.5000

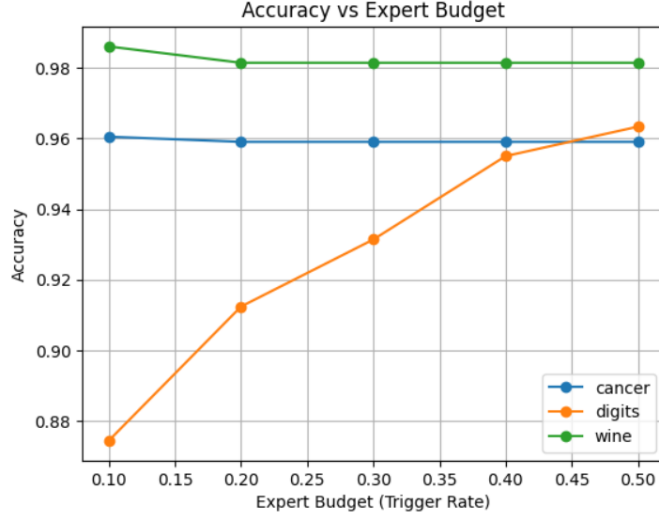


Figure 6: Cross-dataset budget curve across breast cancer, digits, and wine.

A.7 Rare-Event Fraud Benchmark

Table 8: Final rare-event budget-sweep summary on the real credit card fraud benchmark.

Policy / Signal	Balanced Acc.	Pos. Recall	Trigger Rate	Cost
Guard only	0.9286	0.8606	0.0000	1.0000
Always expert	0.8968	0.7939	1.0000	9.0000
Oracle	0.9301	0.8606	0.0050	1.0402
Wobble (best, 0.1% cap)	0.9288	0.8606	0.0006	1.0048
Margin (best)	0.9287	0.8606	0.0001	1.0006
Confidence (best)	0.9287	0.8606	0.0001	1.0008
Combined (best)	0.9286	0.8606	0.0000	1.0002
Random (best)	0.9287	0.8606	0.0202	1.1616

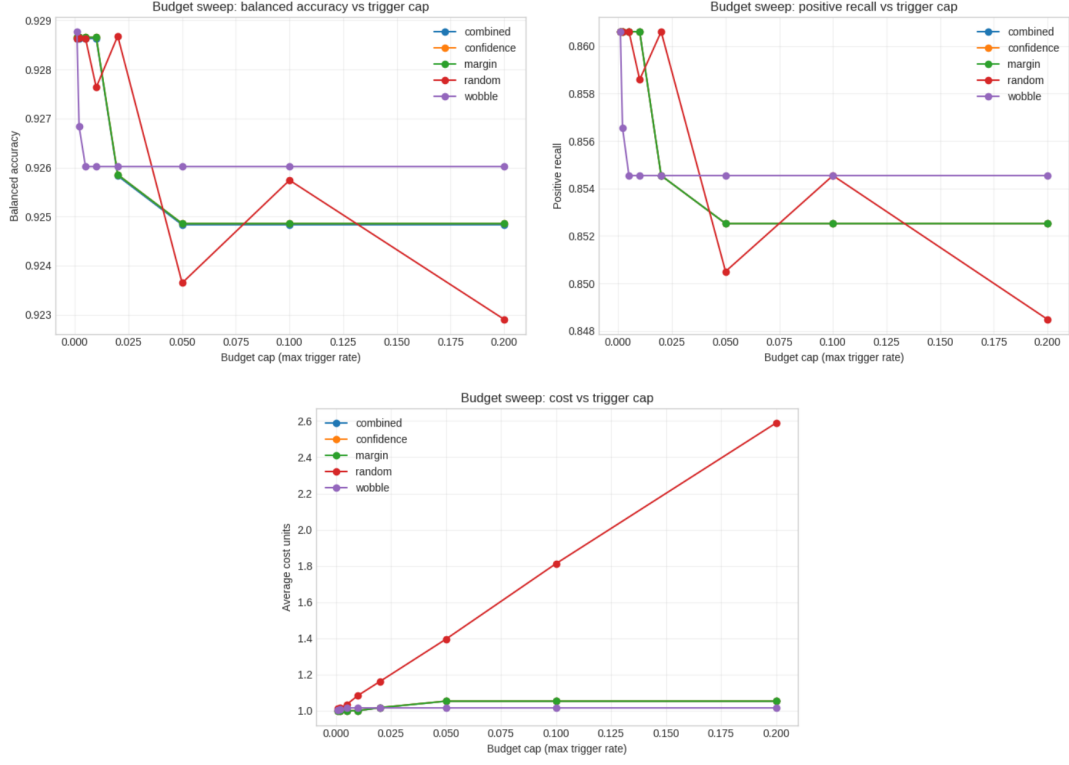


Figure 7: Rare-event budget-sweep plots for balanced accuracy, positive recall, and cost under fixed trigger caps.